# MIT 8.012 Problem Set 4 - Problem 3.16 (K&K)

3.16 Water shoots out of a fire hydrant having nozzle diameter $D$ with nozzle speed $V_0$. What is the reaction force on the hydrant?

Solution (handwritten) follows on the next page.

Problem (3.16) (KAK)

Consider a very small portion of the water

$$\Delta m\, V_0 = \Delta F\, \Delta t \quad \text{(Impulse Equation)}$$

$$\Delta m = \underbrace{\rho}_{\text{Density of water}} \cdot \Delta V = \rho \pi \frac{D^2}{4} V_0 \Delta t$$

$$\Delta V = \pi R^2 \Delta w$$

$$= \pi \frac{D^2}{4} \Delta w$$




$$= \pi \frac{D^2}{4} V_0 \Delta t$$

$$\rho \pi \frac{D^2}{4} V_0^2 \cdot \Delta t = \Delta F \Delta t$$

$$\Delta F = \rho \pi \frac{D^2}{4} V_0^2$$

$$dF = \rho \pi \frac{D^2}{4} V_0^2$$